# Jonathan Andrew Pedroza PhD

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## Education

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|---|---|---|
| University of Oregon | Doctor of Philosophy, Prevention Science<br>*Specialization: Educational Data Science* | 2021 |
| University of Oregon | Master of Science, Prevention Science | 2020 |
| California State University (CSU) Dominguez Hills | Master of Arts, Health Psychology | 2017 |
| Cal Poly Pomona | Bachelor of Arts in Psychology<br>*Minor: Latina/o & Chicana/o Studies* | 2014 |

## Publications

**Pedroza, J. A.**, Guardado-García, J. J., Williams, M. J., Alcala, M., Nakamura, N. N., Serpas, D. G., Harrison, A., Chon, D. (2026). Measurement invariance of the LGBT People of Color Microaggressions Scale among LGBTQ+ Latina/o/x and LGBTQ+ non-Latina/o/x People of Color. *LGBT Health, 13*(5) 1–10. https://doi.org/10.1177/23258292261456304

**Pedroza, J. A.**, Serpas, D. G., Guardado-García, J. J., Zelaya, D. G., Alcala, M., & Keys, K. (2026). Cardiovascular disease risk factors, intersectional minority stressors, and self-reported cardiovascular disease among sexual and gender minority people of color: An exploratory latent class analysis. *LGBT Health, 13*(4) 1–10. https://doi.org/10.1177/23258292261452696

Nichols, L. M., Fleming, C. M., **Pedroza, J. A.**, O'Brien, K. M., & Tanner-Smith, E. E. (2024). Psychometric Properties of the Alabama Parenting Questionnaire Among Adolescents with Substance Use Disorder Histories. *Child and Adolescent Social Work Journal*, 1-14.

Nichols, L. M., **Pedroza, J. A.**, Fleming, C. M., O'Brien, K. M., & Tanner-Smith, E. E. (2021). Social-ecological Predictors of Opioid Use among Adolescents with Histories of Substance Use Disorders. *Frontiers in Psychology*, 12.

Rosen, L. D., Carrier, L. M., **Pedroza, J. A.**, Elias, S., O'Brien, K. M., Lozano, J., … & Ruiz, A. (2018). The Role of Executive Functioning and Technological Anxiety (FOMO) in College Course Performance as Mediated by Technology Usage and Multitasking Habits. *Psicologia Educativa, 24(1)*, 14-25.

*Manuscripts Under Review*

Lopez, J. L. S., **Pedroza, J. A.**, Usovich, K., Van Dusen E., & Norouzi, N. Conceptualizing Programming and Mathematics Self-Concept Among Community College Data Science Students.

Yan, W., Wilkerson, M. H., & **Pedroza, J. A.** K-12 Teachers' Perception and Navigation of Computational Notebooks as Data Investigation Tools.

*Manuscripts In Progress*

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**Pedroza, J. A.** Pychometric: Using Python and Stan for Bayesian Estimated Diagnostic Classification Models.

Updated March 2026

# Presentations

## National Conferences

**Pedroza, J. A.** (September 2020). *Examining the Longitudinal Association Between Access to Recreational Resources and Leisure-time Physical Activity Across California*. Virtual presentation at the American Public Health Association Physical Activity Twitter Conference. (https://twitter.com/PaSectionAPHA/status/1309192269447143425).

Nichols, L. M., Fleming, C. M., **Pedroza, J. A.**, O'Brien, K. M., & Tanner-Smith, E. E. (August 2020). *Psychometric Properties of the Alabama Parenting Questionnaire Among Adolescents in Recovery from Substance Use*. Poster presented at the American Psychological Association Convention. (Virtual Conference).

Nichols, L. M., Fleming, C. M., **Pedroza, J. A.**, O'Brien, K. M., & Tanner-Smith, E. E. (April, 2020). *Psychometric Properties of the Alabama Parenting Questionnaire Among High-risk Adolescents*. Poster to be presented at the Collaborative Perspectives on Addiction Conference, San Diego, CA. (Poster accepted but conference canceled).

Luther, G., **Pedroza, J. A.**, & Kelly, N. R. (November 2019). *The Role of Perceived Racial Discrimination and Weight Status on Depression Among Latina Women*. Poster presented at The Obesity Society, Las Vegas, NV.

**Pedroza, J. A.**, Kelly, N. R., & Budd, E. L. (November 2019). *Neighborhood Walkability is Associated with Physical Activity in Latinx Adults, Particularly Those with High Acculturation*. Oral presentation accepted at the American Public Health Association Conference, Philadelphia, PA.

Cahn, A. F., **Pedroza, J. A.**, Terral, H. & Budd, E. L. (November 2019). *Neighborhood Walkability, Physical Activity, and Sedentary Behaviors Examined by Perceived Health Status among Diverse Latinx Adults*. Oral presentation presented at the American Public Health Association, Philadelphia, PA.

Folger, A. C., **Pedroza, J. A.**, Smith, T. M., Budd, E. L., & Kelly, N. R. (May 2019). The Impact of *Individual-Level Motivators of and Barriers to Healthy Food Choices on Perceived Healthy Food Access in a Rural Community*. Poster accepted for presentation at Society of Prevention Research Conference, San Francisco, California.

**Pedroza, J. A.**, Luther, G. M., Folger, A. C., Smith, T. M., Kelly, N. R., & Budd, E. L. (May 2019). *Smartphone Use as a Moderator of Bicultural Dimensions of Acculturation and Sedentary Behaviors Among a Sample of Latinx Young Adults*. Poster accepted for presentation at Society of Prevention Research Conference, San Francisco, California.

**Pedroza, J. A.**, Folger, A. C., Luther, G. M., & Smith, T. M., Kelly, N. R., & Budd, E. L. (May 2019). *Assessing Nutrition Knowledge in Rural Communities: Examining the Association between Sociodemographic Factors and Nutrition Knowledge*. Poster accepted for presentation at Society of Prevention Research Conference, San Francisco, California.

Folger, A. C., **Pedroza, J. A.**, Smith, T. M., Budd, E. L., & Kelly, N. R. (August 2019). *Depression and Chronic Health Conditions: Examining Racial Identity as a Moderator*. Poster accepted for presentation at American Psychological Association Convention, Chicago, Illinois.

Smith, T. M., **Pedroza, J. A.**, Kashani, Y., Kelly, N. R., Budd, E. L., & Giuliani, N. R. (January 2019). *Depressive Symptoms are Associated with Body Mass Index among Young Adults with Higher Education*. Poster to be presented at National Multicultural Conference and Summit, Denver, Colorado.

**Pedroza, J. A.**, Smith, T. M., & Budd, E. L. (July 2018). *Language Competence and Screen-time Sedentary Behaviors are Moderated by Smartphones*. Poster presentation at APA Division 45 Research Conference, Austin, Texas.

## Regional Conferences

Lopez J. S. L., Usovich, K., Van Dusen E., Norouzi, N., & **Pedroza, J. A.** (October 2026). Interdisciplinary Perspectives for Centering Student Experience and Identity in Data Science Program Evaluation. Poster presentation accepted at the INSPIRE Conference, Los Angeles, CA.

**Pedroza, J. A.**, Kelly N. R., & Budd, E. L. (October 2019). *The Association of Neighborhood Walkability and Physical Activity in Differing Levels of Acculturated Latinx Adults*. Oral presentation accepted at the Oregon Public Health Association Conference, Corvallis, OR.

Lieberman, N., **Pedroza, J. A.**, Elias, S., Jimenez, M., Cheever, N. A., Carrier, L.M., & Rosen, L. D. (April 2017). *The Psychology of New Technologies: Exploring Executive Functioning, Anxiety, Physical Activity, and Time Perception*. Symposium presented at Western Psychological Association Convention, Sacramento, California.

Lim, A. F., **Pedroza, J. A.**, O'Brien, K., Ceja, L., Jimenez, M., Carrier, L.M., & Rosen, L. D. (April 2016). *Using Functional Near-Infrared Spectroscopy to Examine the Impact of Personal Technology Usage on Executive Functioning*. Symposium presented at Western Psychological Association Convention, Long Beach, California.

Lim, A. F., **Pedroza, J. A.**, Jimenez, M., Morales, A., Carrier, L.M., & Rosen, L. D. (April 2016). *New Developments in Language Brokering Research Among Latinas/os*. Symposium presented at Western Psychological Association Convention, Long Beach, California.

**Pedroza, J. A.**, Franco, D., Franco, J. A., & Norman, T. J. (April 2016). *Student Satisfaction in International College Students*. Presentation at the Western Decision Sciences Institute, Las Vegas, Nevada.

**Pedroza, J. A.**, Franco, J. A., Franco, D. & Norman, T. J. (April 2016). *Google Glass: The Influence of Recording Interviews on Prefrontal Cortex Activation*. Presentation at the Western Decision Sciences Institute, Las Vegas, Nevada.

## Local Conferences

*denotes mentored undergraduate students

**Pedroza, J. A.** & Strang A. (March 2026). *Assessing Statistics Understanding: Early Detection of Learning Gaps using Diagnostic Classification Models*. Poster presentation at Berkeley School of Education Research Day, Berkeley, California.

**Pedroza, J. A.**, Yan, W., & Wilkerson, M. H. (March 2026). *Using Computational Notebooks as Teaching Tools: The Show Your Work Mapping Notebook*. Poster presentation at Berkeley School of Education Research Day, Berkeley, California.

Lindsey, M.*, Smith, T.M., & **Pedroza, J. A.** (May 2018). *Latinx Perceptions of Their Environment: The Examination of Neighborhood Walkability by Region in a National Latinx Sample*. Poster presentation at Undergraduate Research Symposium, Eugene, Oregon.

Bender, C.*, Smith, T.M., & **Pedroza, J. A.** (May 2018). *Examining Risk and Protective Factors in the Relationship Between Discrimination Stress and Body Mass Index (BMI) in a Sample of Latinx Adults*. Poster presentation at Undergraduate Research Symposium, Eugene, Oregon.

**Pedroza, J. A.** & Smith, T. M. (May 2018). *The Examination of Smartphone Use as a Moderator on the Association between Acculturation and Sedentary Behaviors*. Poster presentation at Graduate Research Forum, Eugene, Oregon.

Smith, T. M., Vargas, K., & **Pedroza, J. A.** (February 2018). *Your Health Matters! Understanding Health Inequities in the Latinx Community*. Invited Presentation at Raza Unida Youth Conference, Eugene, Oregon.

**Pedroza, J. A.**, Carrier, L.M., & Rosen, L. D. (February 2017). *The Effects of Media and Technology Use on Psychological Well-Being Independent of Health Habits*. Presentation at Student Research Day, Carson, California.

**Pedroza, J. A.** & Rosen, L. D. (February 2017). *Pokémon GO and Meet People. Examination of Technology's Effect on Physical Activity and Social Connectedness*. Presentation at Student Research Day, Carson, California.

**Pedroza, J. A.** & Rosen, L. D. (February 2016). *Technology and Mental Well-being: A Literature Review Examining Media Use and Health Factors' Effects on Mental Health*. Poster presentation at Student Research Day, Carson, California.

**Pedroza, J. A.** & Morales, A. (March 2014). *Latinos Living with Diabetes: A Content Analysis of Interventions for Self-Management*. Poster presentation at Student Research Conference, Pomona, California.

## Research Experience

UC Berkeley — Fall 2026 - Present
PI: Michelle H Wilkerson PhD — GitHub: Show Your Work
*Developing Pedagogical Computing Practices for K-12 Educators Using Computational Notebooks*

- Conducted 10 think-aloud interviews with reflective questions for educators on using computational notebooks for teaching spatial analysis
- Converted 4 notebooks from Python to R while retaining computational and pedagogical efficiency for both languages
- Examined educators' interest, confidence, and barriers to using computational notebooks

UC Berkeley — Fall 2026 - Present
PI: Alexander Strang PhD — GitHub: TODO CLEAN
*Assessing Internal Motivators & Developing Methods to Make Data Science More Equitable*

- Developed a workflow to clean quiz data, run a Bayesian Diagnostic Classification Model in Stan, and output a parameterized report with model diagnostics, findings, and explanations, as well as a list of proficiencies in concepts assessed for each student
- Creating an app for instructors to input data and custom priors and model type and output a report
- Co-developed a proxy measure to evaluate student perceived readiness for statistics and data science courses

CSU Fullerton — Fall 2025 - Present
PI: James J Guardado-García PhD
*Exploring the Role of Intersectional Minority Stress in LGBTQ+ People of Color*

- Validated the LGBTQ People of Color Microaggressions Scale using confirmatory factor analysis to assess measurement invariance among Latina/o and non-Latina/o People of Color
- Tested key cardiovascular risk factors to identify different groups of varying risk level using Latent Class Analysis among LGBTQ+ People of Color

University of Oregon — Fall 2019 - Summer 2024
PI: Emily E. Tanner-Smith PhD
*Examining the Roles of Social-Ecological Predictors Among Adolescents With Substance Use Disorders*

- Validated both an adolescent and parent reported parenting questionnaire which was linked to developmental, behavioral and mental health outcomes
- Established an ecological approach to detect associations from intrapersonal, interpersonal, school, and communal factors with opioid use in at-risk adolescents

University of Oregon — Spring 2019 - Summer 2021
PIs: Elizabeth L Budd PhD, MPH & Nichole R Kelly PhD
*Addressing Health Disparities in Latina/o/x Adults*

- Investigated individual- and community-level determinants of health, including discrimination stress, perceived health status, and nutrition knowledge in underserved Latina/o/x populations
- Analyzed national and regional datasets to examine how built environment factors (e.g., neighborhood walkability) and sociocultural factors (e.g., acculturation, language competence) influence physical activity, screen-time, and sedentary behaviors among Latina/o/x adults

University of Oregon — Fall 2017 - Fall 2018
PI: Tasia S Smith
*Understanding Food Access in Rural Communities*

- Built relationships with key community members to have locations for survey data collection and host focus groups
- Trained several research assistants on how to conduct community-based research, including conducting a needs assessment, organizing data collection, and analyzing quantitative and qualitative data
- Mentored research assistants' projects where they presented at conference presentations and one resulted in an undergraduate thesis

CSU Dominguez Hills — Spring 2015 - Summer 2017
PIs: Larry D Rosen PhD & L Mark Carrier PhD

*Exploring the Role of Technology Use on Health and Cognition*

- Trained various research assistants on how to prepare cognitive assessments and fit a functional near-infrared spectroscopy (fNIRS) device on high- and low-technology use participants
- Led lab workshops on research design, presentation skills, preparing for graduate school, and statistical analyses
- Developed workflow to clean fNIRS data and match with behavioral and cognitive data for various projects

CSU Dominguez Hills Spring 2017 - Summer 2017
PI: Michelle Martinez EdD
*McNair Scholars Program*

- Created presentations for graduate school preparation
- Taught students how to create personal statements, diversity, statements, and Curriculum Vitaes
- Provided edits on graduate school materials
- Assisted scholars with literature review, hypothesis formulation, methodology, analyses, and the writing process

Cal Poly Pomona Fall 2013 - Summer 2017
PI: Alejandro Morales PhD
*Investigating Prefrontal Cortex Activation in Adult Latina/o/x Language Brokers*

- Led experiments where young adults would interpret conversations between a medical provider and family members

# Teaching Experience

## Instructor

UC Berkeley Fall 2026
DATA H195A: *Data Science Honors Thesis Seminar* Class Materials

- Accepted 24 data science honors applicants as part of the honors admissions committee
- Mentor students to develop independent research projects to be presented on research methods, academic writing, and reproducibility

UC Berkeley Fall 2026
DATA 6: *Introduction to Computational Thinking with Data Science and Society* Class Materials

- Advise a graduate teaching assistant on improving teaching methodologies during discussions
- Adopt workflow utilizing GitHub, Jupyter, and Typst to build handwritten and coding assignments
- Develop data science assignments to practice the fundamentals of coding while also developing critical thinking through the application of these skills using real-world data

Cal Poly Pomona Spring 2022
PSY 4433/A: *Experimental Psychology & Activity* Class Materials

- Trained 30 students to create independent research projects where they conducted literature reviews, developed research questions and hypotheses, incorporated theoretical frameworks, utilized

reliable and valid measures, designed experimental stimuli, conducted inferential statistics, and wrote academic reports
- Created unique approaches to collect data and introduce experimental stimuli from student participation pools during an online course

Cal Poly Pomona — Fall 2021 - Spring 2022
PSY 3307/A: *Statistics for Psychology & Activity* — Class Materials

- Taught multiple sections of 30+ students on fundamental statistical tests for Psychology and other social sciences along with learning considerations for research applications
- Instructed students on how to calculate statistics by hand, and using SPSS and R

### Teaching Assistant

University of Oregon — Winter 2020
FHS 420: *Research in Human Services*
Instructor: Justin Caouette PhD

University of Oregon — Fall 2019
FHS 328: *Human Development in the Family Context*
Instructor: Bertranna Muruthi PhD

CSU Dominguez Hills — Spring 2017
UNV 395: *Special Topics, McNair Scholars Program*
Instructor: Michelle Martinez EdD

CSU Dominguez Hills — Spring 2016
PSY 415 & 416: *Advanced Research Methods in Cognitive Psychology*
Instructor: Alex Lim MA

CSU Dominguez Hills — Fall 2015
PSY 235: *Introduction to Research Methods*
Instructor: Alex Lim MA

Cal Poly Pomona — Fall 2014
PSY 455: *Human Sexual Behavior: Relationships*
Instructor: Alejandro Morales PhD

## Invited Talks

UC Berkeley — Fall 2026
*SESAME Colloquium*
Topic: Examining Student Internal Motivators and Ways For Improving Competency in an Accelerated Statistics Course

University of Oregon — Winter 2025
PREV 607: *Graduate Seminar*
Topic: Careers for Prevention Scientists

Instructor: Ashley Linden-Carmichael PhD

University of Oregon — Winter 2021
FHS 492: *Contemporary Issues in Public Health*
Topic: Community-based COVID-19 Prevention & Control Research
Instructor: Elizabeth L Budd PhD, MPH

University of Oregon — Fall 2020
PREV 633: *Contemporary Issues in Public Health*
Topic: Community-based COVID-19 Prevention & Control Research
Instructor: Elizabeth L Budd PhD, MPH

University of Oregon — Fall 2018
FHS 215: *Exploring Family and Human Services*
Topic: Working with Low Income Rural Populations
Instructor: Kavitha Rao MEd, MA

University of Oregon — Winter 2018
FHS 420: *Research in Human Services*
Topic: Experiences Using fNIRS Neuroimaging Techniques
Instructor: Emily Tanner-Smith PhD

## Relevant Experiences

Posit - Posit Academy — Jan 2022 - Present
Data Science Mentor
Supervisor: Kristin Bott

- Lead collaborative and presentation sessions in R and Python for cohorts ranging from research institutions to financial institutions
- Mentor learners on fundamentals of R and Python, like visualizations, data wrangling, data analysis, dashboards, and report creation
- Propose changes for curriculum to align with course learning objectives

University of Kansas - Accessible Teaching, Learning, & Assessment Systems — Sep 2022 - Jul 2024
Data Scientist
W Jake Thompson PhD

- Took the lead as the co-chair of the Diversity, Equity, Inclusivity, and Belonging Committee
- Developed automated, parameterized reports to present current findings and compare trends to previous years for collaborating institutions to disseminate on their websites
- Delivered rapid ad-hoc analyses to support center-wide research initiatives
- Constructed predictive models evaluating the text-to-speech functionality to improve accessibility for diagnostic models assessments
- Advanced Dynamic Learning Maps (DLM) assessments, by developing models to assess linear dependencies between attributes
- Built interactive dashboards for state-level stakeholders to identify school districts with challenges implementing DLM assessments

- Utilized natural language processing to improve response rate to teachers implementing DLM assessments

University of Oregon - Student Corps to Combat Coronavirus — Jun 2020 - Sep 2020
Team Lead Contact Tracer
Jeffrey Measelle PhD

- Collaborated with the local public health department to track cases and contacts
- Interviewed potential positive cases of COVID-19
- Supervised several contact tracers
- Provided resources for contacts to get necessities during quarantine

### Diversity-Related Experiences

UC Berkeley - D-Lab — Jan - May 2026
Latinx AI Fellowship Mentor

- Shared personal academic trajectories with mentees
- Discussed how my culture influences my research
- Discussed mentees' research interest and the benefits and drawbacks of pursuing a graduate degree

University of Oregon - Division of Graduate Studies — Sep 2020 - Jun 2021
Diversity & Inclusion Graduate Employee

- Created advisory groups to meet with departments to improve inclusivity for the Division of Graduate Studies
- Cleaned data for departments to choose candidates for Promising Scholar awards
- Organized data from McNair Scholars programs to match with graduate programs at the university
- Assisted students address with departmental issues during office hours
- Hosted professional development sessions with graduate students from various disciplines

## Awards

### Scholarships & Fellowships

CSU Chancellor's Doctoral Incentive Program — 2020-2021
General University Scholarship - $2,700 — Fall 2018, Fall 2020
Graduate Research Advancement & Development Grant — Spring 2016, Spring 2017
Promising Scholar Award — Fall 2017
Graduate Equity Fellowship - $5,000 — Summer 2016

### Travel Awards

INSPIRE 2026 Travel Scholarship — Fall 2026
Society of Prevention Research ECPN Travel Award — Spring 2019
APA Division 45 Conference Travel Award — Summer 2018
American Psychological Association's Travel Award — Spring 2018

### Presentation Awards

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| Runner up Best Student Oral Presentation Award | American Public Health Association Conference - Physical Activity Section | 2019 |
| 1st & 2nd Place Oral Presentation | CSU Dominguez Hills Student Research Day | 2017 |
| Winning Paper and Poster Presentation | South Bay Economic Forecast Conference | 2016 |
| 1st Place Oral Presentation | CSU Dominguez Hills Student Research Day | 2016 |
| 1st Place Oral Presentation | CSU Dominguez Hills Student Research Day | 2016 |

## Training & Service

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| International Society of the Learning Sciences Reviewer | 2026 |
| American Public Health Association Physical Activity Section Abstract Reviewer | 2020 |
| American Psychological Association Advanced Training Institute – Research Methods with Diverse and Ethnic Group | 2017 |

## Package Development Contributions

- ThresholdR GitHub Repository

## Open-Source Projects

### Dashboards

- Dashboard on Graduate Student Exit Survey Data From (Docker Image)
- Dashboard of the Importance of Graduate Student Teaching and Undergraduate Credit Hours

### Web Apps

- App for Assessing Student Competencies
- App for Building a Discrete Bayesian Network (Docker Image)
- Percentile Ranking for Psychological Assessments